

15.439 Quantitative Investment Management

Assignment 2

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1 Abstract / Executive memo

Conclusions. We use PERMNO for stable firm identity; Price_Ret(T1) vs PRC differ when dividends occur (65,599 rows match, 24,123 differ; 138 companies have Price_Ret(T1) $\equiv$ PRC in every month within 0.0001). Constituent count 500–506; 795 unique PERMNOs, 285 in all 180 months; mcap percentiles by date and at exit/entry are computed. Fama–MacBeth with skip-month momentum PRC_Ret(T12M1) yields a **positive but insignificant** mean coefficient for F1M (mean ≈ 0.0023 , $t \approx 0.44$, $p \approx 0.66$); the “November 2019” regression uses the return that realizes in November (Oct 31–Nov 30), i.e. cross-section as of October 2019 EOM, and is negative. No univariate or bivariate predictor is significant; F3M/F6M need Newey–West for overlap. Log(mcap) weighting gives similar results. **Recommendation:** Momentum [-12,-1] does not show significant cross-sectional predictability here; use F1M for inference; if using F3M/F6M, report HAC standard errors.

2 Data, identifiers, and price vs total return

Q1. Why do we use PERMNO instead of TICKERS?

PERMNO is CRSP’s permanent identifier and is stable across corporate actions (splits, name and ticker changes); ticker is the exchange symbol and can be reused or changed. For panel data and cross-database linking, PERMNO avoids ambiguity and tracks the same firm through time.

Q2. Price_Ret(T1) vs PRC: (a) Is the difference a data error? (b) When are they equal?

We define Price_Ret(T1) = $(P_t/P_{t-1}) - 1$ by permno; PRC is total return (price plus reinvested dividends). (a) The difference is **not** a data error: PRC includes dividends and adjustments; they differ when dividends or distributions occur. (b) They are equal only in no-dividend months. In our sample: 65,599 rows match (within 0.0001), 24,123 differ; 138 companies have Price_Ret(T1) $\equiv$ PRC in every month. We save Price_Ret(T1) for later use.

3 Company counts and market-cap percentiles

Q3. How many companies per date? Is it always 500?

Constituent count per date ranges from 500 to 506 (median 503). This is not a mistake: the S&P 500 is reconstituted periodically, and timing of additions/deletions can yield slightly more than 500 names in some months.

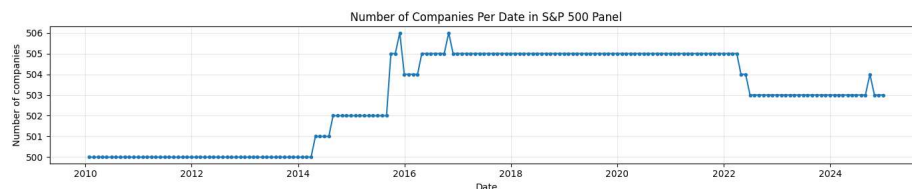


Figure 1: Number of companies per date in the S&P 500 panel.

Q4. How many unique PERMNOs and how many are in the sample every month?

There are **795** unique PERMNOs and **285** companies present in all 180 months.

Q5. Percentile set for market cap by date; also month-before-exit and month-of-entry.

For each month, we compute the full cross-sectional percentile set of market capitalization for all S&P 500 constituents, visualizing the time series of these percentiles to track distribution changes over time. In addition,

we calculate and plot the same set of percentiles specifically for (a) stocks in the month prior to their exit from the index, and (b) stocks in the month they first enter. This approach allows us to analyze how the market cap profiles of entrants and exiting companies compare with the overall panel distribution. The table below summarizes the median value of each percentile set across the sample period, providing a concise numerical reference. Detailed plots of these percentile time series are available in Appendix 9.3, where the three market cap percentile figures are shown together for comparison.

Table 1: Median of each percentile set, Values in USD Billions

Percentile	Main time series	Month prior to exit	Month of entry
Min	2.62	4.50	12.49
5th	5.55	4.83	12.74
25th	11.12	6.23	13.60
Median	20.00	6.53	14.25
75th	40.33	8.79	16.95
95th	174.17	11.13	17.10
Max	774.29	11.41	177.34

4 Trailing and forward returns; percentile plots

Q6. Construction of trailing and forward returns; percentile plots.

All return calculations are based on geometric compounding. For clarity, consider the general trailing k -month return formula for stock i at date t :

$$R_{i,t}^{(k)} = \prod_{s=0}^{k-1} (1 + r_{i,t-s}) - 1$$

where $r_{i,t}$ is the monthly return (including or excluding dividends as appropriate). We compute various trailing and forward windows, such as the full year trailing ($k = 12$), trailing 11 months (skipping the most recent), or forward multi-month horizons.

These values are summarized each month via cross-sectional percentiles to trace distributional shifts over time. The table below offers a compact snapshot of the median percentile values, illustrating both the skewness and broader distribution dynamics (such as the wide interquartile range).

For full interpretive context—including detailed time series plots and the code for calculation—please see the Jupyter notebook.

Table 2: Median of each percentile set for each variable

Variable	Min	25th	Median	75th	Max
Price_Ret_T1	-0.376	-0.022	0.010	0.109	0.361
PRC_Ret_T12	-0.595	-0.021	0.122	0.272	1.499
Prices_Ret_T12	-0.745	-0.055	0.089	0.241	3.957
PRC_Ret_T12M1	-0.571	-0.018	0.110	0.246	1.353
Prices_Ret_T12M1	-0.748	-0.049	0.084	0.223	3.660
PRC_Ret_T12_1M	-0.236	-0.021	0.012	0.048	0.310
Prices_Ret_T12_1M	-0.356	-0.022	0.011	0.044	0.336
Vol_Prices_Ret_T12M1	0.023	0.055	0.071	0.091	0.926
SR_Prices_Ret_T12M1	-6.420	-0.697	1.317	3.821	16.787
PRC_Ret_F1M	-0.252	-0.021	0.012	0.048	0.348
PRC_Ret_F3M	-0.374	-0.027	0.036	0.102	0.560
PRC_Ret_F6M	-0.490	-0.034	0.068	0.162	0.810

5 Fama–MacBeth: momentum and one-month forward return (Q7)

Methodology. We use the two-step Fama–MacBeth procedure.

1. Each month t , we run a cross-sectional regression of forward return on predictor(s): $R_{i,t}^{\text{fwd}} = \gamma_{0,t} + \gamma_t x_{i,t} + \varepsilon_{i,t}$ (OLS; in Q10 we use WLS with weights $w_i = \log(\text{mcap}_i)$).

2. We collect the time series $\{\gamma_t\}$ and test the mean coefficient $\bar{\gamma} = \frac{1}{T} \sum_t \gamma_t$ with $t = \bar{\gamma}/\widehat{SE}$ and $\widehat{SE} = \widehat{\sigma}(\gamma_t)/\sqrt{T}$. Months with fewer than 30 valid stocks are dropped.

Q7(a). November 2019 cross-sectional regression.

Methodology. Single cross-section at one date (October 2019 EOM): y is the one-month forward return (Oct 31–Nov 30 2019); x is skip-month momentum (e.g. PRC_Ret(T12M1)). Same OLS as one “month” of the FM procedure.

Coefficient	R^2
−0.053	0.052

The negative coefficient in that month indicates short-term reversal: higher skip-month momentum was associated with lower next-month return. This is one month only; we do not infer the full sample from it.

Q7(b). Full-sample Fama–MacBeth.

Methodology. Full FM: same cross-sectional regression every month; we obtain the coefficient series $\{\gamma_t\}$ and apply time-series inference (mean, SE, t , p) as above.

Results. Over 168 months: mean coefficient ≈ 0.00225 , $t \approx 0.44$, $p \approx 0.66$ —**not significant**. The coefficient series fluctuates and flips sign (e.g. post-2008, 2020).

Interpretation. The point estimate is positive but small and insignificant; the momentum premium is not stable over time. Coefficients fluctuate and frequently change sign, illustrating that the momentum premium is not stable over time (e.g. reversals around 2015, 2020).

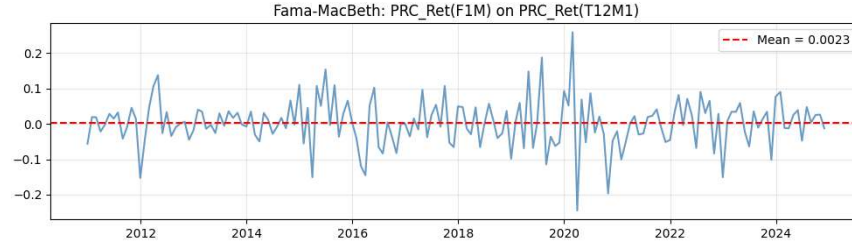


Figure 2: Time series of Fama–MacBeth coefficient γ_t (PRC_Ret(F1M) on PRC_Ret(T12M1)) by month.

Q7(c). Interpretation for an educated layperson.

The conclusion follows from the full-sample FM inference and the time series of coefficients. Economically, a positive mean would support momentum; here we do not find reliable predictability. In plain terms: stocks that did well recently (excluding the last month) do not show a consistent tendency to do better next month in this sample. Results are not consistent across time—there are notable periods of sign change. The plot of γ_t highlights variation and reversals (e.g. post-2015, 2020).

Q7(d). Performance measures (Peterson Ch.2).

Methodology. We treat the monthly coefficient series $\{\gamma_t\}$ as a “strategy return” series and compute hit rate, max drawdown, and Sortino ratio (downside deviation).

Hit rate	Avg Gain	Avg Loss	Slugging Ratio	Max DD	Sortino	Ann Sharpe	Ann Sortino
51%	0.050	−0.047	1.050	−0.653	0.034	0.117	0.118

Interpretation. Each month’s coefficient corresponds to the expected excess return of a long-high/short-low momentum portfolio per unit exposure; treating the series as strategy returns yields the metrics above. A hit rate near 50%, deep max drawdown (≈ -0.65), and low Sortino/Sharpe indicate that the implied premium is unstable and not attractive from a risk-adjusted perspective. Figure 3 shows sustained drawdowns (e.g.

2017–2019, 2021–2024) and a coefficient distribution centered near zero and roughly symmetric—consistent with no reliable momentum edge in this sample. See Appendix 9.10 for the computation.

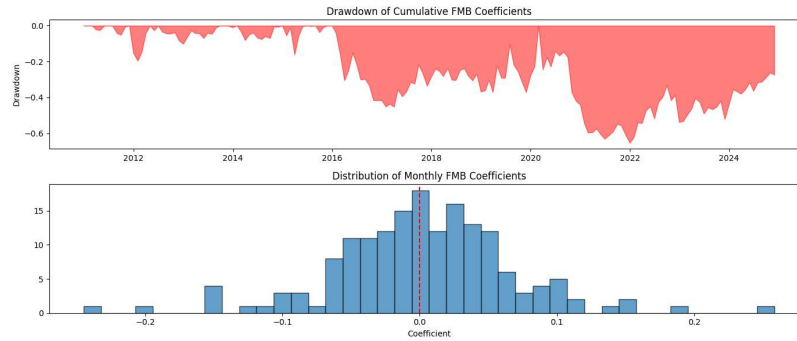


Figure 3: Performance measures treating the FMB coefficient series as strategy returns (Q7d)

6 Univariate and multivariate Fama–MacBeth (Q8)

Methodology. Same Fama–MacBeth two-step as in Q7. We first run univariate regressions (one predictor at a time); univariate specs ignore correlation across predictors. We then run four bivariate specifications; bivariate gives conditional effects (holding the other regressor constant) but multicollinearity can inflate standard errors and flip signs versus univariate.

Q8(a). Univariate Fama–MacBeth.

Results. Table 5 (Appendix) reports mean coefficient, t -stat, p -value, and number of dates for each of the nine predictors (e.g. last-month return, total return, trailing 12-month and skip-month momentum, single-month-at- t -12, volatility-adjusted momentum).

Interpretation. None of the nine predictors has a significant mean coefficient ($p > 0.05$ for all). Means are small and mixed in sign—reversal and momentum effects can pull in opposite directions. Univariate results do not control for other factors.

Q8(b). Multivariate (bivariate) Fama–MacBeth.

Methodology. Four bivariate regressions:

1. Last-month return and skip-month momentum
2. Total return and skip-month momentum
3. Total return and trailing 12-month price return
4. Total return and volatility-adjusted skip-month momentum (SR)

Same FM inference per coefficient.

Results. Results are in Appendix 9.2. Mean coefficients and t -stats for the four specs are similar in spirit to univariate but with two regressors the interpretation is “holding the other constant.”

Interpretation. Economically, one might expect momentum predictors to have a positive sign (higher past performance $\Rightarrow$ higher future return) and short-term reversal (e.g. last month return) to have a negative sign. With two regressors, multicollinearity can inflate SEs and signs may flip versus univariate; each coefficient is the marginal effect holding the other constant.

7 Forward 3- and 6-month returns (Q9)

Methodology. Same Fama–MacBeth structure as in 7(b), but the dependent variable y is now the forward 3-month or 6-month return (PRC_Ret(F3M), PRC_Ret(F6M)); predictor remains PRC_Ret(T12M1).

Consecutive months share most of the return window, so γ_t is autocorrelated and the usual standard error is too small (inflated t). We correct using Newey–West HAC standard errors with max lags $L = 6$: $\widehat{\text{Var}}(\bar{\gamma})$ accounts for autocorrelation up to L lags (Bartlett kernel).

Results. Table 3 reports mean coefficient and inference for F3M and F6M (F1M as in 7(b) is the reference).

Table 3: Fama–MacBeth: F1M (ref), F3M, F6M; usual and Newey–West t , p .

Forward return	Mean coef	t (usual)	t (NW)	p (usual)	p (NW)
F3M	0.0056	0.7010	0.4654	0.4843	0.6423
F6M	0.0107	0.8962	0.4515	0.3715	0.6522

Interpretation. F3M and F6M point estimates are larger than F1M but still **not significant**. After Newey–West correction, t -stats drop (overlap is the main econometric issue). Alternatives: non-overlapping forward returns (e.g. F3M every 3 months) or block bootstrap.

8 Weighted regression: log(market cap) (Q10)

Methodology. We repeat the Fama–MacBeth procedure of 7(b) but weight each observation by $\log(\text{mcap})$ (WLS). We minimize $\sum_i w_i (R_i - \gamma_0 - \gamma x_i)^2$ with $w_i = \log(\text{mcap}_i)$. The weighted coefficient is the “typical” effect with more weight on larger-cap stocks.

Results. Table 4 shows: equal-weight mean coef 0.00225 (t 0.44, p 0.66); $\log(\text{mcap})$ -weight mean coef 0.00242 (t 0.48, p 0.63)—slightly higher but still insignificant (additional figures in Appendix 9.5).

Weighting	Mean coef	t -stat	p -value
Equal	0.00225	0.44	0.66
$\log(\text{mcap})$	0.00242	0.48	0.63

Table 4: Equal vs. $\log(\text{market cap})$ weighted Fama–MacBeth: mean coefficient, t -stat, and p -value for PRC_Ret(T12M1) predicting one-month forward return.

Interpretation. $\log(\text{mcap})$ weighting is relevant for a cap-weighted index perspective and dampens the influence of noisy small firms. Equal-weight OLS can be dominated by small caps. Comparing equal vs. $\log(\text{mcap})$: similar mean coefficient, still insignificant. Caveats: endogeneity (size–momentum); the weighted coefficient is the typical effect for the weighted observation, not the equal-weight average; $\log$ dampens extremes versus raw mcap .

9 First principles (Q11)

Q11. What does this assignment teach (first principles)?

- **What we did:** Data construction (identifiers, total vs. price return, panel structure), momentum signals (trailing returns, skip-month T12M1, volatility-adjusted SR), and Fama–MacBeth cross-sectional prediction (one-month and multi-month forward returns, equal and weighted).
- **What we learned:** Different dependent variables (F1M vs. F3M/F6M) and different weighting (equal vs. $\log(\text{mcap})$) change inference; overlapping forward returns induce autocorrelation in γ_t , so Newey–West HAC or non-overlapping samples are needed.
- **Takeaway:** Building a factor, testing it with Fama–MacBeth, and interpreting results with appropriate caveats (overlap, weighting, multicollinearity in multivariate specs).

Appendix

9.1 Univariate Fama–MacBeth results (Q8a)

Table 5: Univariate Fama–MacBeth: one-month forward return on each predictor. Mean coefficient, t -stat, p -value, and number of dates.

Predictor	Mean coef	t -stat	p -value	N dates
Price_Ret(T1)	0.0011	0.08	0.93	178
PRC	−0.0053	−0.37	0.71	179
PRC_Ret(T12)	0.0019	0.37	0.71	168
Prices_Ret(T12)	−0.0008	−0.23	0.82	167
PRC_Ret(T12M1)	0.0023	0.44	0.66	168
Prices_Ret(T12M1)	0.0007	0.21	0.84	167
PRC_Ret(T12_1M)	−0.0040	−0.37	0.71	167
Prices_Ret(T12_1M)	−0.0037	−0.38	0.70	166
SR_Prices_Ret(T12M1)	0.0001	0.30	0.77	167

9.2 Multivariate Fama–MacBeth results (Q8b)

Mean coefficients and t -stats for the four multivariate specifications; see Table 6 below.

Table 6: Multivariate Fama–MacBeth: mean coefficient, t -stat, p -value for each specification.

Specification	Predictor	Mean coef	t -stat	p -value
Price_Ret_T1+Prices_Ret_T12M1	Price_Ret_T1	0.0027	0.2179	0.8277
Price_Ret_T1+Prices_Ret_T12M1	Prices_Ret_T12M1	0.0017	0.5198	0.6038
PRC+PRC_Ret_T12M1	PRC	−0.0011	−0.0794	0.9367
PRC+PRC_Ret_T12M1	PRC_Ret_T12M1	0.0040	0.8751	0.3827
PRC+Prices_Ret_T12	PRC	−0.0012	−0.0858	0.9316
PRC+Prices_Ret_T12	Prices_Ret_T12	0.0007	0.2488	0.8037
PRC+SR_Prices_Ret_T12M1	PRC	−0.0001	−0.0135	0.9892
PRC+SR_Prices_Ret_T12M1	SR_Prices_Ret_T12M1	0.0002	0.6911	0.4904

9.3 Market cap percentile plots

All percentiles trend upward; dispersion between min/max and central percentiles widens over time. Notable dips (e.g. early 2020) align with market stress.

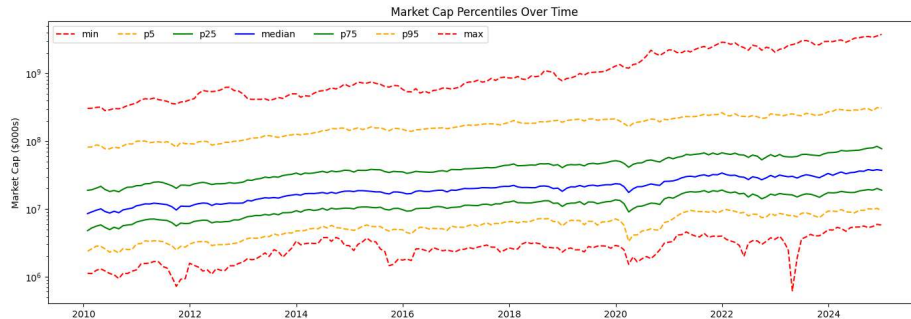


Figure 4: Market cap percentiles over time for all S&P 500 constituents.

Distribution of exiting firms is volatile and cyclical; percentiles move together, with periods of higher vs. lower market caps before exit.

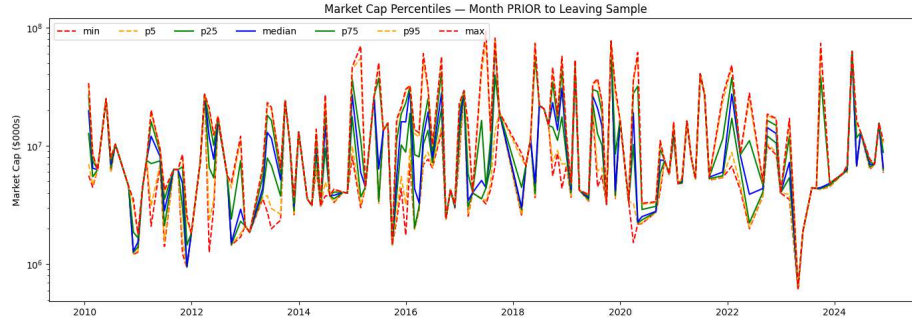


Figure 5: Market cap percentiles for stocks in the month *prior* to leaving the sample.

Entering firms show a wide range of sizes (min to max) and time variation in the size distribution of new constituents.

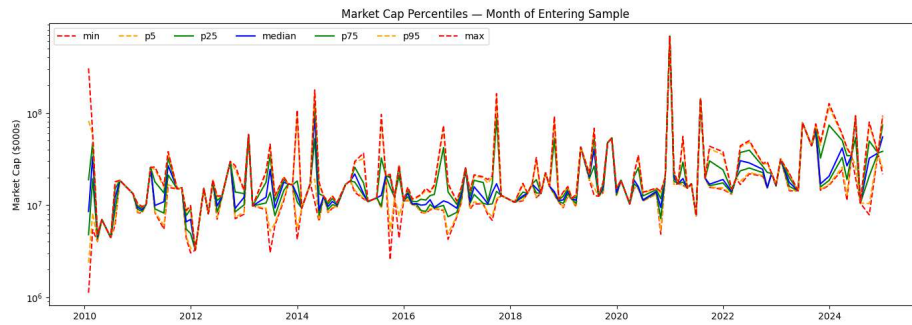


Figure 6: Market cap percentiles for stocks in the month they *enter* the sample.

9.4 Plots of Fama-MacBeth coefficients for varied univariate regressions

The subplots in Figure 7 show that none of the predictors have a mean coefficient far from zero; coefficients are volatile and often change sign, consistent with no significant univariate predictability.

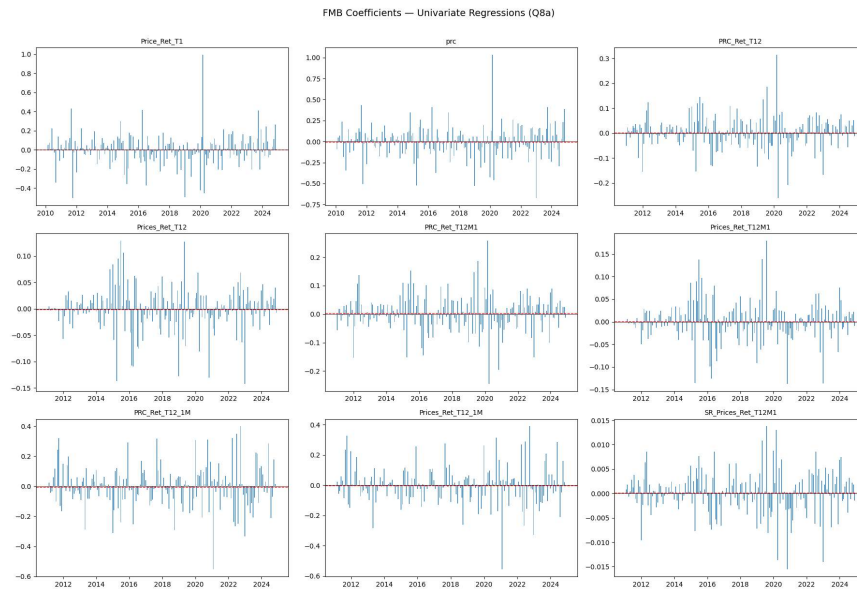


Figure 7: Fama-MacBeth coefficient time series for each of the nine univariate predictors (Q8a).

9.5 Plots of Fama-MacBeth coefficients for weighted vs unweighted

The top chart in Figure 8 plots monthly coefficient by date; the two series are nearly identical. The bottom one plots cumulative coefficient; weighted and unweighted paths track each other, so log(mcap) weighting does not materially change the estimated premium or its time-series behavior.

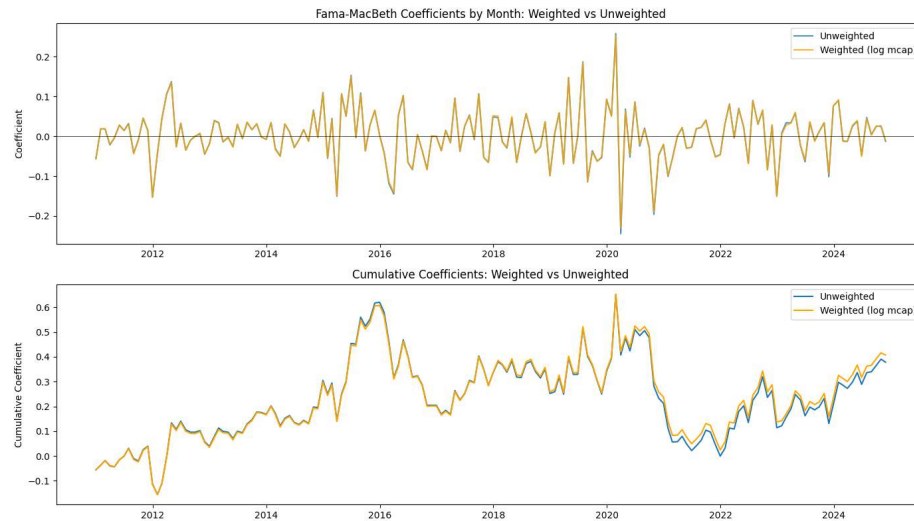


Figure 8: Comparison of unweighted (OLS) and log(mcap)-weighted (WLS) Fama-MacBeth coefficients.

9.6 Reproducibility

Run all cells of HW2/assignment2.ipynb, with SP500Raw.xlsx in same directory or path set in the first cell. Outputs: in-notebook figures and tables.

9.7 Relevant code (build full panel)

From assignment2.ipynb (Helpers cell): geometric trailing and forward returns.

Listing 1: build_full_panel (excerpt)

```

1 def build_full_panel(df):
2     out = df.sort_values(["permno", "date"]).copy()
3     g = out.groupby("permno", group_keys=False)
4     out["Price_Ret_T1"] = g["price"].pct_change()
5     # PRC_Ret_T12: (1+prc).rolling(12).apply(prod)-1 per permno
6     # PRC_Ret_T12M1: 11 months t-12..t-2, product of (1+prc) shifted by 1
7     # PRC_Ret_F1M = g["prc"].shift(-1); F3M, F6M = product of next 3/6 months
8     out["log_mcap"] = np.log(out["mcap"].clip(lower=1e-6))
9     return out

```

9.8 Fama-MacBeth (cross-section and inference)

From assignment2.ipynb (Notebook Helpers): OLS per date, then time-series inference.

Listing 2: run_fama_macbeth and fama_macbeth_inference

```

1 def run_fama_macbeth(panel, y_col, x_cols, weight_col=None, ...):
2     for date, g in panel.groupby(date_col):
3         if valid.sum() < min_obs: continue
4         res = cross_sectional_ols(y, X, weights)
5         rows.append({"coef": res, "_date": date})
6     return pd.DataFrame(rows).set_index("_date")
7

```



```

8 def fama_macbeth_inference(coef_series):
9     mean, std = coef_series.mean(), coef_series.std()
10    t_stat = mean / (std / sqrt(T))
11    p_value = 2 * (1 - t.cdf(abs(t_stat), T-1))
12    return {"mean": mean, "t_stat": t_stat, "p_value": p_value, ...}

```

9.9 Newey–West HAC (Q9)

From assignment2.ipynb: HAC variance for the mean of the coefficient series (autocorrelation up to L lags, Bartlett kernel). Used for overlapping forward returns (F3M, F6M).

Listing 3: Newey–West t -stat for mean coefficient

```

1 def newey_west_t_stat(coef_series, max_lags=6):
2     # HAC variance: Var(mean) with Bartlett kernel, lags 1..L
3     # t_nw = mean / sqrt(Var_nw(mean))
4     ...

```

9.10 Strategy metrics (Q7d)

From assignment2.ipynb: Hit rate, avg gain, avg loss, slugging ratio, max drawdown, sortino, annualized sharpe, annualized sortino.

Listing 4: Strategy metrics

```

1 def strategy_metrics(coef_series):
2     # Minimal version: all metrics in a few lines
3     coefs = coef_series.dropna()
4     if coefs.empty:
5         return {k: np.nan for k in ["hit_rate", "avg_gain", "avg_loss", "slugging", "max_dd",
6                                     "sortino", "sharpe", "annualized_sortino"]}
7     hit_rate = (coefs > 0).mean()
8     avg_gain, avg_loss = coefs[coefs > 0].mean() if (coefs > 0).any() else np.nan, coefs[
9     coefs < 0].mean() if (coefs < 0).any() else np.nan
10    slugging = abs(avg_gain / avg_loss) if (not np.isnan(avg_loss) and avg_loss != 0) else
11    np.inf
12    max_dd = (drawdown := (cum := coefs.cumsum()) - cum.cummax()).min()
13    downside_dev = np.sqrt((coefs[coefs < 0]**2).mean()) if (coefs < 0).any() else np.nan
14    m, s = coefs.mean(), coefs.std()
15    sortino = (m * 12) / (downside_dev * np.sqrt(12)) if downside_dev and downside_dev > 0
16    else np.nan
17    sharpe = (m * 12) / (s * np.sqrt(12)) if s > 0 else np.nan
18    return dict(hit_rate=hit_rate, avg_gain=avg_gain, avg_loss=avg_loss, slugging=slugging,
19               max_dd=max_dd, sortino=sortino, sharpe=sharpe, annualized_sortino=sortino)

```